

ML4P Problem Set 2

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Github link:<https://github.com/VM2708/ML4P/tree/main/Desktop/ML4P/ps2>

1 Problem 1: Warm Up

1.1 Proof of symmetry in the architecture

My architecture will be as follows. I will go from a single input to a linear layer of 128 with weights that can be trained, a nonlinear activation in the form of nn.RELU, and then another linear layer with learnable weights that bring the data back to one final point. The forward pass in my architecture takes the absolute value of the input, making the problem even. Mathematically, this looks as follows:

$$h_{\theta}(x) = W_2 * \sigma(W_1|x| + b1) + b2$$

. For the proof of symmetry, $h_{\theta}(-x) = W_2 * \sigma(W_1|-x| + b1) + b2 = W_2 * \sigma(W_1|x| + b1) + b2 = h_{\theta}(x)$

1.2 Comparing Symmetrized and Nonsymmetrized architecture

Table 1 shows that the symmetrized architecture consistently outperforms the non symmetrized architecture for the various even functions. Figure 1 shows one example comparing the residuals for both symmetrized and non-symetrized architecture.

Function	Symmetrized L1 Loss	Non Symmetrized L1 loss
cos(x)	0.0079	0.0152
x^2	0.2716	0.6518
x^4	69.9173	90.8667
cos(x)	0.2205	1.2655

Table 1: L1 Loss comparisons for both symmetrized and non symmetrized architecture

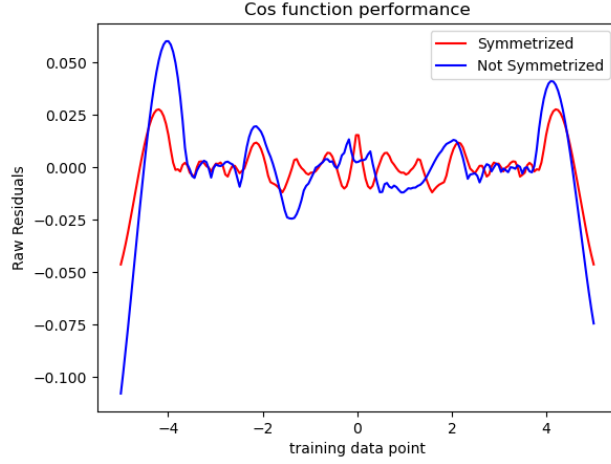


Figure 1: Example of the residuals plotted for symmetrized versus non-symmetrized architecture

2 Problem 2: Designing Symmetry-Respecting Architectures

2.1 Part 1: Choose a Symmetry and Task

2.1.1 Description of dataset

I will be using the "turbulent radiative layer 2D" dataset from *The Well* for this problem. I mainly chose this dataset because it used the smallest storage space. This dataset describes the motion of cold, dense gas on the bottom and hot, dilute gas on top. Once a Kelvin-Helmholtz instability is introduced, the induced turbulence causes a net mass flux from the hot phase to the cold phase. This process recreates the mass transfer between cold clouds in the galaxy that pass through the ambient, hot medium.

Boundary Conditions: Periodic in the x-direction, zero-gradient for the y-direction (which means that there is no flux going in or out in the y-direction) Dimensions of data: 101 timesteps of 384x128 images. Fields available: Density, pressure, and velocity fields (v_x and v_y). Number of data points: 90 simulations/trajectories

2.1.2 Symmetries being enforced

The symmetry I will be enforcing here is the periodicity in the x-direction. Shifting the field horizontally would change where the boundary point is located but should not change the physics of how the system evolves over time.

2.2 Part 2

2.2.1 Construction

In order to enforce the defined symmetries, I have defined the following block:

$$\text{ConvBlock}(x) = \sigma_{\text{GELU}}(\text{GroupNorm}(\text{PeriodicConv}(x)))$$

This is where the symmetrization is enforced. The Periodic convolution layer is equivariant, and uses 3x3 blocks where there is circular periodicity in the x direction and no wrapping in the y direction.

This is used for the following ML architecture where e is the encoder, b is the next bottleneck, d is the decoder, and $\phi \in \mathbb{R}^{4*H*W}$ is the inputted data.

$$e_1 = \text{ConvBlock}(\phi) \in \mathbb{R}^{16*H*W}$$

$$b = \text{ConvBlock}(\text{MaxPool}(e_1)) \in \mathbb{R}^{32*H/2*W/2}$$

$$d_1 = \text{ConvBlock}(\text{ConvTranspose}(b)) \in \mathbb{R}^{16*H*W}$$

$$\hat{\phi} = \text{ConvBlock}_{1x1}(d_1) \in \mathbb{R}^{4*H*W}$$

The **MaxPool** is a version of local pooling for each block. There will not be any global pooling, since this is not a classification problem.

2.2.2 Proof of Concept

My architecture satisfies translational equivariance in the x-direction for shifts that are multiples of the pooling stride. This holds because:

PeriodicConv2d uses circular padding in x, so convolution commutes with cyclic shifts — shifting the input then convolving gives the same result as convolving then shifting.

MaxPool2d and ConvTranspose2d are both spatially local operations that do not depend on absolute x-position, so they commute with shifts that are multiples of their size (2 and 3 respectively).

GroupNorm and GELU operate elementwise on channel values and are position-independent, so they trivially commute with any spatial shift.

The composition of all these layers would then satisfy translational equivariance as well.

2.3 Part 3

2.3.1 Implementation

In the Code.ipynb file.

2.3.2 Sanity Checks

I checked to make sure the size of the output file is correct and that the equivariance model produces the same result when the data is shifted, whereas the baseline should produce a different result. I also plotted one example of the evaluated data vs the actual data. For both the equivariant model and the baseline, these images were not very accurate, which could be attributed to the small number of epochs and subset of data used for the training.

2.3.3 Comparison

Metric	Equivariant Model	Baseline
Validation VRMSE	35.1507	31.1331
Training Time (s)	298.216	193.693449974
Test VRMSE	32.1689	27.256

Table 2: Caption

None of my training tricks broke symmetry. My optimizer (Adam), my use of Group Norm (which does not mix spatial information), and my activator (GELU applied elementwise in space) do not affect the spatial information and therefore do not break symmetry. I did not use any other tricks. Testing with random seeds resulted in pretty similar VRMSE values, proving that this method is robust to multiple seeds.

2.3.4 Interpretation

Honestly, my baseline does outperform my equivariant model. It has a lower VRMSE (variance-scaled root mean squared error) and takes significantly less time to train. However, I trained my models with a limited dataset and few Epochs in order to allow for time constraints (honestly, I got tired of waiting for the code to run). The limited dataset could have contributed to neither models reaching the point where they best described the data. I also used very simple architecture (with one encoder and decoder) for ease of understanding how the models were defined. Both of these choices could have led to the models not being able to work to their true potential.